

# Day 06 - GitHub Remotes & Collaboration

## Topics Covered

Remote Repositories, Origin & Upstream, git init vs git clone, Remote Commands, Remote Tracking Branches, GitHub Flow, Pull Requests, Branch Protection, Merge Conflicts, Non-Fast-Forward Push, Production Collaboration Workflow.

## Important Commands

```
git remote
git remote -v
git remote add origin <url>
git fetch
git pull
git pull --rebase
git push
git push -u origin main
```

## Remote Workflow

Local Repository → git push → GitHub Repository  
GitHub Repository → git fetch/git pull → Local Repository

## Origin vs Upstream

origin = your GitHub repository.  
upstream = original repository in a fork workflow.

## GitHub Flow

main → feature branch → commit → push → Pull Request → Code Review → CI → Merge

## Merge Conflict

Resolve conflict markers manually, then run:

```
git add .
git commit
git push
```

## Non-Fast-Forward

If push is rejected:

```
git pull --rebase
git push
```

## Branch Protection

Require Pull Requests, approvals, status checks, and prevent direct pushes to main.

## Key Takeaways

git fetch only downloads changes. git pull downloads and integrates them. Remote-tracking branches (origin/main) are updated by fetch. Professional teams collaborate through feature branches, pull requests, reviews, and branch protection.